$0x1 + 0x2 + 0x4 + 0x8 + 0x10 + 0x20 = 0x3f$

[0x33]

$2^6 - 1 \rightarrow$ single bit ??

parity

is popcnt(rax) % 2 → 0-bit(rax)

is $x_{n+1} = x_n \oplus (x_n >> 2^n)$

is let $b = (b_{63}, \ldots, b_0) \in V_2 = \mathbb{F}_2^{64}$ sign bit ?? 0

where $\mathbb{F}_2 = GF(2)$,

$x \leftarrow x \oplus (x >> 2^k)$

$k = 0, 1, 2, 3, 4, 5$

$\phi(b) = \sum_{i=0}^{63} b_i \pmod 2 = 1^T b$

augmentation map $\mathbb{F}_2^{64} \rightarrow \mathbb{F}_2$

popcnt(x) mod 2

total parity

```
mov     rdx, rax        rdx = rax_0
shr     rdx, 0x1        rdx = rax_0 >> 1
xor     rax, rdx        rax = rax_0 ⊕ (rax_0 >>
                        rdx =
mov     rdx, rax
shr     rdx, 0x2        rdx = (rax_0 ⊕ (rax_0 >> 1))
xor     rax, rdx        rax = ((rax_0 ⊕ (rax_0 >> 1)) ⊕ (rax_0 ⊕ (rax_0 >> 1)))
                        rax = rdx
mov     rdx, rax
shr     rdx, 0x4
xor     rax, rdx
```

rdx = ((rax ⊕ (rax_0 >> 1)) ⊕ (rax_0 ⊕ (rax_0 >> 1))) >> 2) >> 4

(rule → 0-bit ← lesson)

0 → 0

1 → 1

0010 → 1

0100 → 1

0101 → 0

0110 → 0

0101 → 1

1011 → 1

1000 → 0

1110 → 0

1111 → 1

00010000 → 0

00010001 → 0

00010010 → 0

00010011 → 1

```
mov     rdx, rax
shr     rdx, 0x8
xor     rax, rdx

mov     rdx, rax
shr     rdx, 0x10
xor     rax, rdx

mov     rdx, rax
shr     rdx, 0x20
xor     rax, rdx
```

see 0x33.asm where I leave copy-paste to see time

(ffff → aaaa, eeee → b4b4, dddd → e6e6, cccc → 8888)

13 ones = 26

ff → aa & ee → b4 & dd → 96

1111 1111 → 1010 1010 ⊕ 1110 1110 → 1011 0100 ⊕ 1101 1101 → 1001 0110 ⊕

6 1's, 20 1's → 0

6 1's, 20 1's → 0

x01 → 1

x11 → 0

x21 → 0

x31 → 1

$x_{n+1} = x_n \oplus (x_n >> 2^n)$

popcnt if n = 0

[0x33]